

Corporate Fare Strategy, Technical Realignment, and Local Operational Blueprint for the Pakyaw Platform

1. Executive Summary & Strategic Positioning (Tricycle-First Market Entry)

The primary bottleneck for early-stage digital transport marketplaces in the Philippines is the complex and highly centralized regulatory structure of the Land Transportation Franchising and Regulatory Board (LTFRB). Transportation Network Vehicle Service (TNVS) licenses for private four-wheel vehicles are bound to strict, regional "slot-based" quotas. Securing these slots and obtaining a Certificate of Public Convenience (CPC) can take months, which can trap a pre-seed startup in regulatory limbo before it can even launch.

To de-risk this venture for investors of the Ormoc Chamber of Commerce and Industry (OrCham), the founder of the Pakyaw platform has designed a **Phase 1 market entry focusing strictly on already-franchised motorized tricycles within Ormoc City**.

Under the Local Government Code of 1991 (Republic Act No. 7160), the franchising authority over motorized tricycles-for-hire was devolved from the national government to Local Government Units (LGUs). Tricycle operators in Ormoc City are licensed locally through the Sangguniang Panlungsod's Motorized Tricycle Operator's Permit (MTOP).

By launching Phase 1 as an on-demand booking tool for already-licensed, independent LGU-franchised tricycle operators, the proponent ensures Pakyaw:

- Bypasses the national LTFRB's TNVS slot waitlists and the immediate requirement for a PHP 10 Million corporate paid-up capital.
- Integrates with a ready-made local fleet of already-legal transit operators.
- Directly addresses the major pain points of Ormoc's commuters (selective drivers, long wait times at terminals, and unmetered fare overcharging).

The platform operates in two distinct modes:

1. **Pakyaw (Solo/Exclusive):** The passenger buys out the vehicle's capacity for a private, direct point-to-point journey.
2. **Share (Carpool):** Multiple unrelated passengers heading in the same direction pool their rides, sharing the fare while the platform dynamically coordinates their boarding sequences.

2. Zonal Taripa Architecture (Ordinance No. 121, S. 2023)

Tricycle passenger fares in Ormoc City are strictly regulated by municipal ordinances. In Ormoc City, tricycle fares are governed by the officially certified **Ordinance No. 121, Series of 2023 (The New Taripa)**, which amends Article III of Ordinance No. 034, Series of 1992.

Under this ordinance, the city is divided into **four (4) distinct fare zones** centered around the **City Stage (City Center)**. Fares are structured based on the destination barangay's designated zone, with strict legal mandates for regular passenger rates, 20% privilege discounts for Senior Citizens, PWDs, and students, and flat-rate surcharges for Special and Night Trips [Image 2].

A. Base Fare Rules within the 2.5km Radius

- **Regular Passengers:** PHP 10.00 flat fare for the first 2.5 kilometers, and an additional PHP 1.50 for every succeeding kilometer.
- **Privileged Passengers (Senior Citizens, PWDs, & Students):** PHP 8.00 flat fare for the first 2.5 kilometers (applying the legislated 20% discount) .
- **Enclosed Barangays (Within 2.5km Radius):** North, South, East, West, Alegria, Bantigue, Camp Downes, Can-adieng, Cogon, Don Felipe Larrazabal, Linao, Punta, and Toog [Image 2].

B. Special Trips Surcharge

- Defined as "trips plying beyond the regular or usual route of the public utility vehicles (PUVs), such as but not limited to entering private subdivisions and private properties." [Image 2]

- **Mandated Surcharge:** An additional flat fare of **PHP 5.00** on top of the regular fare [Image 2].

C. Night Trips Surcharge

- Defined as "trips from nine o'clock in the evening (9:00 p.m.) until five o'clock in the morning (5:00 a.m.)." [Image 2]
- **Within 2.5km Radius:** An additional flat fare of **PHP 5.00** on top of the regular fare [Image 2].
- **Outside 2.5km Radius:** An additional flat fare of **PHP 10.00** on top of the regular fare [Image 2].

D. Official Barangay Zonal Tariff Schedule [Image 1, Image 2, Image 3]

Zone	Barangay	Regular Rate (PHP)	Senior, PWD & Student (PHP)
Zone 1 - Red / Freezone	Brgy. North, East, West, South	10.00	8.00
	Alegria, Bantigue, Batuan, Camp Downes, Can-adieng, Cogon, Don Felipe Larrazabal, Linao, Punta, Toog	10.00	8.00
	Doña Feliza Mejia	11.00	8.80
	Naungan	12.00	9.60

	San Isidro, Tambulid	12.00	9.60
	Bagong Buhay / Bliss	13.00	10.40
	San Pablo	14.00	11.20
	Alta Vista	15.00	12.00
	Libertad, Patag, Sto. Niño, San Antonio	16.00	12.80
	Salvacion	19.00	15.20
	Donghol	20.00	16.00
	Airport, Nasunogan	22.00	17.60
Zone 2 - Blue	Ipil, Panali-an	13.00	10.40
	Danhug, Macabug, Sumangga	17.00	13.60
	Can-untog	19.00	15.20

	Mabini	25.00	20.00
Zone 3 - Yellow	Dayhagan	18.00	14.40
	Juaton	19.00	15.20
	Concepcion	21.00	16.80
	Luna	22.00	17.60
	Cabulihan	23.00	18.40
	Cagbuhangin	24.00	19.20
	Dolores, Valencia	25.00	20.00
	Milagro	29.00	23.20
	San Jose	31.00	24.80
	Matica-a	33.00	26.40
	Sabang Bao	35.00	28.00
	Hibunaon	36.00	28.80
Zone 4 - Green	Lilo-an	19.00	15.20

	Licama	22.00	17.60
	Curva, Lao, Tzu Chi	23.00	18.40
	Labrador, Margen, San Vicente	27.00	21.60
	R.M. Tan, San Juan	31.00	24.80
	Mas-In	32.00	25.60
	Nueva Sociedad	37.00	29.60
	Green Valley	40.00	32.00
	Leondoni	41.00	32.80
	Esperanza	43.00	34.40
	Manlilinao	49.00	39.20

3. Resolving the "Tricycle Tariff Trap": The Pakyaw (Solo) Buyout Strategy

If a platform strictly applies the LGU's literal definition of a "Special Trip" (which allows

only a PHP 5.00 flat surcharge on top of the regular single fare) to a private on-demand solo ride, a passenger would only pay the base fare plus PHP 5.00 (e.g., PHP 15.00 total for Zone 1) [Image 2].

If an app forces a driver to accept an exclusive point-to-point booking for only PHP 15.00, **the driver will immediately reject the booking or uninstall the app**. They are giving up the chance to load 5 other passengers on the street. Traditionally in Ormoc, drivers expect a **full vehicle capacity buyout (traditionally calculated as 6 seats)**

when hired for a private point-to-point journey (e.g., 15 PHP $\times$ 6 seats = PHP 90.00 total).

Conversely, forcing a single passenger to pay for all 6 physical seats is too expensive for daily commuters.

The Solution: The Capacity-Based Buyout Algorithm

To satisfy both the LGU's legal framework and the driver's economic requirements, the platform implements a **Dynamic Capacity Buyout for the Pakyaw (Solo) mode**. Instead of charging for a flat 6 seats or a static 4-seat cap, the **4-seat buyout acts strictly as a Minimum Floor (Lower Bound)** for exclusivity. The base fare dynamically scales up to the full 6-seat capacity if the booking passenger inputs a larger group headcount ($N_{\text{passengers}}$). Fares are split on the receipt into the Base Buyout Fare (passed 100% to the driver) and a separate, legally distinct Platform Technology Convenience Fee (F_{tech}):

$$Fare_{\text{Solo}} = (T_{\text{zone}}(B) \times \max(4, N_{\text{passengers}})) + S_{\text{surcharge}} + F_{\text{tech}}$$

Where:

- $T_{\text{zone}}(B)$ = The mandated regular tariff for destination barangay B under Ordinance No. 121 [Image 1].
- $N_{\text{passengers}}$ = The number of passengers boarding the vehicle ($1 \leq N_{\text{passengers}} \leq 6$).
- $S_{\text{surcharge}}$ = LGU surcharges: Special Trip PHP 5.00 [Image 2], or Night Surcharge

(PHP 5.00 to PHP 10.00) [Image 2].

- F_{tech} = Flat Platform Convenience Fee (set at **PHP 15.00**).

Use Case 1: The Solo Shopper (1 Passenger Booked Alone)

Rider 1 is shopping alone at Gaisano and wants an exclusive private ride back to **Barangay Camp Downes**. They book Pakyaw with **1 passenger**:

- $T_{\text{zone}}(\text{Camp Downes}) = \text{PHP } 10.00$ (Regular Rate) [Image 3]
- $N_{\text{passengers}} = 1$ (Triggers the **4-seat minimum floor**: $\max(4, 1) = 4$)
- $S_{\text{special}} = \text{PHP } 5.00$ (Mandated Special Surcharge for door-to-door drops) [Image 2]
- $F_{\text{tech}} = \text{PHP } 15.00$

$$Fare_{\text{Solo}} = (10.00 \times 4) + 5.00 + 15.00 = \text{PHP } 60.00$$

*The shopper pays PHP 60.00 for private door-to-door convenience (cheaper than the predatory PHP 90.00–120.00 street special rates). The driver collects PHP 45.00 legal fare plus their 60% share of the platform fee (PHP 9.00), earning **PHP 54.00 net**.*

Use Case 2: The Group Outing (6 Passengers Booked Together)

Rider 1 wants to book an exclusive Pakyaw ride from the downtown terminal to **Barangay Camp Downes** for themselves and 5 friends (totaling **6 passengers**):

- $T_{\text{zone}}(\text{Camp Downes}) = \text{PHP } 10.00$ [Image 3]
- $N_{\text{passengers}} = 6$ (Scales to the full 6-seat rate: $\max(4, 6) = 6$)
- $S_{\text{special}} = \text{PHP } 5.00$ [Image 2]
- $F_{\text{tech}} = \text{PHP } 15.00$

$$Fare_{\text{Solo}} = (10.00 \times 6) + 5.00 + 15.00 = \text{PHP } 80.00$$

*The group pays PHP 80.00 total. Split among the 6 friends, it is only **PHP 13.33 per head** for an on-demand private ride. The driver takes home **PHP 74.00 net** on a*

single run.

4. Zero-Bleed Carpooling Strategy: The TODA-Zoned Virtual Corridor Stop Heuristic

Operating a completely unstructured, dynamic, door-to-door carpool route with random, on-the-fly stops across Ormoc is an operational and financial failure. It causes massive, unpredictable detour delays, burns excess fuel, and causes drivers to reject bookings.

To eliminate platform cash bleed and protect driver wages, the founder completely banned on-the-street hailing in "Share" mode. Instead, the platform implements a **TODA-Zoned Virtual Corridor Stop Heuristic**:

—————→	(Direct Route)	—————→
Rider 1, 2, 3 Board	No Detours,	Riders Alight
Simultaneously	No Fuel Wasted	

How the Algorithm Works

1. **Spatio-Temporal Epoch Queue:** When Passenger A requests a "Share" ride from Camp Downes to Mejia, the request is placed into an online **3-minute batching window** (epoch).
2. **Linear Corridor Route Vector:** The system plots a straight, direct route along the main highway corridor (Palo–Carigara–Ormoc Road) toward Mejia, with **zero backtracking permitted**.
3. **Landmark-Locked "Virtual Stops" (Random Pickup Mitigation):** To prevent drivers from wasting fuel entering deep, narrow interior alleys, the algorithm restricts pickups and drop-offs strictly to pre-approved **Virtual Stops** (prominent barangay landmarks, e.g., Camp Downes Tanod Outpost, Cogon Combado

Chapel). Passengers walk a short distance (under 200 meters) to these designated stops to board.

4. **Spatio-Temporal Search Radius Constraints:** To ensure mathematical validity and eliminate illogical detour routing, the matching algorithm enforces two strict spatial boundaries:
 - R_p (**Passenger Walking Radius**) ≤ 200 **meters:** The maximum distance a passenger is routed to walk to a Virtual Stop.
 - R_d (**Driver Detour Radius**) ≤ 500 **meters:** The maximum perpendicular distance the vehicle can deviate from the primary route vector.
5. **Opposite-Coast Matching Prevention:** Camp Downes (South) and Punta (Northwest) lie on opposite ends of Ormoc Bay. By enforcing the $R_d \leq 500m$ constraint, a driver running a route from Camp Downes to Mejia will **never** be matched with a passenger in Punta. This prevents fuel-wasting zig-zag detours across the city center.
6. **The Occupancy Lock (Zero-Bleed):** The app **will not dispatch a tricycle until at least three (3) passengers** along the exact same corridor route are matched.
7. **The Zero-Bleed Fallback:** If the 3-minute timer hits zero and the 3-passenger minimum is not met, **no driver is dispatched, and the platform burns zero subsidy cash.** The system prompts the passenger: *"Your carpool line did not fill. Would you like to keep waiting, or instantly upgrade to a Solo Pakyaw ride to get a driver immediately?"*.

Use Case 3: Neighborhood-to-Neighborhood Matched Run (Camp Downes to Brgy. Mejia)

Commuting from Barangay Camp Downes to Barangay Doña Feliza Z. Mejia traditionally costs PHP 21.00 because passengers must take one tricycle to the public market (PHP 10.00), transfer, and take another tricycle to Mejia (PHP 11.00) [Image 3].

Using the Pakyaw platform:

- **Passenger 1** (Camp Downes $\rightarrow$ Mejia regular tariff ₱11.00) books "Share" from the *Camp Downes Outpost Stop*. Upfront price quoted: **PHP 16.00** (₱11.00 base

fare + ₱5.00 shared tech fee).

- **Passenger 2** (Cogon Combado → Mejia regular tariff ₱11.00) books "Share" from the *Cogon Combado Stop* [Image 3]. Upfront price: **PHP 16.00**.
- **Passenger 3** (Don Felipe Larrazabal → Mejia regular tariff ₱11.00) books "Share" from the *Larrazabal Stop*. Upfront price: **PHP 16.00**.
- **The Match & Dispatch:** Since all three are waiting at Virtual Stops along the same direct linear corridor (Palo–Carigara–Ormoc Road), the app matches them, dispatches a tricycle, and they board sequentially as the vehicle moves in a straight line.

Why This Works on the Ground:

- **The Passengers Win:** Every passenger enjoys a direct, transfer-free, on-demand ride for only PHP 16.00, saving PHP 5.00 over the traditional two-ride cost.
- **The Driver Wins (Brutal Math):** The driver collects the three fares (Total PHP 48.00). After deducting Pakyaw's 40% share of the convenience fees ($3 \times \text{₱}2.00 = \text{₱}6.00$), the driver takes home **PHP 42.00 in net earnings on a single, continuous, non-backtracking run**.

5. Operational Safeguards: Eradicating "Off-App" Bookings and Platform Leakage

A major operational risk is **platform leakage** (or "off-app booking"). This occurs when a driver on a pooled trip attempts to pick up cash-paying, street-hailing passengers who did not use the app, delaying the original passenger and robbing the platform of revenue.

Pakyaw prevents this using three distinct tech-driven safeguards:

A. Headless Linux AI Server & Real-Time GPS Anomaly Detection

The startup's central office houses a localized AI Server running geohashing and trajectory models.

1. **Stop Anomaly Detection:** The server continuously monitors the driver's real-time GPS telemetry stream. If the tricycle stops for more than 45 seconds at any

coordinate not designated as an app-pickup point, the system triggers an automatic "Mismatched Stop" alert to the admin dashboard.

2. **Path Deviation Detection:** If the vehicle deviates from the assigned route vector by more than 150 meters to hunt for street passengers, the system flags a "Path Deviation Anomaly".

B. App-Locked Seat Capacity & Passenger-Policed Compliance

In "Share" mode, the passenger app explicitly displays the active, app-registered seat count: *"Your ride is pooled. Current seats booked: 3 of 6. Do not pay for unlisted passengers."*

- If a driver attempts to load offline passengers, the original passenger can tap an in-app **"Report Seat Abuse"** button.
- This automatically refunds the passenger's convenience fee and docks the driver's weekly completed-ride incentive score, creating a strong community-policed compliance mechanism.

C. Laminated QR Code Driver-Vehicle Pairing

To manage driver shifts under the traditional "boundary" system and prevent unauthorized vehicle use, the platform enforces an **Automated Driver-Vehicle Pairing** mechanism:

1. **The Owner's Only Job:** The fleet owner registers the tricycle's OR/CR and MTOP franchise once on the dashboard, generating a unique, high-security QR code sticker.
2. **Physical Placement:** This QR code is printed as a durable, weatherproof sticker and placed inside the sidecar.
3. **Shift-Level Pairing:** At the start of a shift, the boundary driver must scan this QR code using their Driver App to go online. This locks their personal driver profile and LTO professional license to that specific vehicle for the duration of their shift.

6. The "QR-Coded Street-Hail Onboarding" Heuristic (Bypassing Bypasses)

To satisfy the driver's desire to maximize seat occupancy on a partially filled "Share"

trip without introducing unregulated, unmonitored street hailing (which leads to platform leakage, safety risks, and passenger delays), the platform integrates street passengers digitally through the **Laminated QR Code Onboarding Heuristic**:

1. **The Scenario:** A tricycle driver is currently running an active, matched "Share" trip with 3 app-paying passengers (leaving 3 vacant seats in their 6-seater vehicle). A commuter on the street waves them down.
2. **The Physical Action:** The driver stops. Instead of negotiating a cash fare under the table (which leads to platform leakage), the driver points to the high-security laminated QR code sticker pasted inside the passenger cab.
3. **The Instant Scanning Portal:** The street passenger scans the QR code with their smartphone. This automatically launches a lightweight, instant-booking web interface on their browser (requiring no native app download or registration at the roadside).
4. **The Algorithmic Compatibility Check:** The passenger inputs their desired drop-off barangay. The central server instantly checks their destination against the driver's active trajectory vector using the **Backtracking and Detour Constraints**:
 - **The Backtracking Constraint:** The new passenger's drop-off point must not cause the driver to backtrack (traverse nodes they have already visited), preventing a bad experience for existing passengers.
 - **The Detour Constraint:** The new passenger's drop-off point must add less than **180 seconds** of travel time to the original passengers' estimated time of arrival (ETA).
5. **The Real-Time Resolution:**
 - **If Approved:** The server instantly registers the street passenger in the database for safety and insurance coverage, displays the exact, non-negotiable LGU zonal fare (e.g., PHP 10.00), and adds a flat PHP 5.00 dynamic convenience fee. The driver's active manifest is updated, and the platform securely captures its commission.
 - **If Rejected:** The server blocks the transaction and alerts the driver's app: *"Route Mismatch. Picking up this passenger adds a 10-minute detour. Booking rejected to preserve existing passenger ETAs."*

7. Diversifying Revenue: High-Margin Provincial

Ecosystem

Relying strictly on a 15% commission from low-cost tricycle rides is difficult to scale. A 15% cut on a PHP 15.00 zonal fare is only PHP 2.25. To gain a strong competitive edge in Ormoc City, the founder will monetize the transit network through three high-margin B2B channels: (replaced nani sa tech fee)

A. The B2B Destination-Based Merchant Referral Model ("Pakyaw Perks")

Instead of trying to make money solely off the commuter's travel fare, **monetize the destination.**

1. **Merchant Partnership:** Pakyaw will partner with local commercial landmarks, supermarkets, hotels, and cafés in Ormoc City (e.g., SM Center Ormoc, Robinsons Place, Sabin Resort, 88 Café).
2. **Digital Coupon Integration:** When a passenger enters one of these partner establishments as their drop-off point, the app automatically generates a digital discount coupon (e.g., *"10% off your purchase at 88 Café"* or *"Free drink at Robinsons food hall"*) on their electronic receipt.
3. **The Revenue Model:** The partner merchant pays the platform a **fixed referral or lead-generation fee** (e.g., PHP 20.00 to PHP 50.00) for every paying customer the transport network physically delivers to their doors. This shifts the business from a low-margin transport company to a high-margin local commerce and ad-tech engine.

B. B2B Logistics and Errand Services (Pasabuy, Pasugo)

Tricycle drivers experience significant idle time during off-peak mid-day lulls (10:00 AM to 3:00 PM). Pakyaw will repurpose this tricycle fleet to handle hyperlocal deliveries and errand services:

- **"Pasabuy" & "Pasugo" (Errand & Concierge Services):** Users pay a premium fee to have a driver do their grocery shopping, pick up dry cleaning, or fetch prescription medicines. Fares use a base delivery fare (e.g., PHP 49.00) plus a per-kilometer surcharge.

DICT PEMEDES Compliance & Amnesty Strategy:

As the logistics vertical scales, it will fall under the jurisdiction of the Department of Information and Communications Technology (DICT). Private Express and/or Messengerial Delivery Service (PEMEDES) rules require regional operators to have a minimum capitalization of PHP 300,000.00. To bypass early capital constraints legally, Pakyaw will leverage the **DICT's ongoing Amnesty Program for unregistered delivery operators**. This program allows the logistics vertical to operate legally, regularizing operations without retroactive penalties while the platform scales.

C. SaaS Fleet Owner Subscriptions (Freemium Model)

Local tricycle operators often manage "boundary" fleets of 5 to 50 vehicles, but struggle with manual tracking, driver shifts, and asset protection.

- The founder offers fleet owners a **freemium SaaS dashboard** to track their units on a live map, monitor daily mileage, and automate driver-vehicle pairing via QR codes.
- While basic tracking is free, advanced features—such as automated maintenance alerts (*"Tricycle #4 has traveled 1,000 kilometers this month; it is time for an oil change"*), automated daily boundary logging, and direct access to high-paying B2B logistics contracts—will require a monthly subscription fee of **PHP 499.00 per fleet**.

8. Financial Optimization & Real-World Sustainability Analysis

To present a credible, institution-grade investment case to the OrCham investors, **this document formalizes the total pre-seed raise at PHP 4,019,500.00**.

A. Surgical Asset Reclassification (COA Circular No. 2022-004)

The initial costs sheet categorized several low-value assets under "CapEx" (Capital Expenditures) to be depreciated over time. Under Philippine accounting guidelines (COA Circular No. 2022-004 and BIR standards), the **capitalization threshold for Property, Plant, and Equipment (PPE) is PHP 50,000.00**. Any tangible asset purchased below PHP 50,000.00 must be classified as **Semi-Expendable Property**

and expensed outright as OpEx.

By applying this legal rule, the founder has moved **16 out of the 19 original CapEx items** from CapEx to OpEx (Semi-Expendable Expenses):

Asset Description	Quantity	Unit Price (PHP)	Total Cost (PHP)	Correct Accounting Category
iOS iPhones	2	45,000	90,000	Semi-Expendable Communication Equipment
UPS & Backup Generator	1	45,000	45,000	Semi-Expendable Machinery and Equipment
Laptops (WFH Setup)	10	35,000	350,000	Semi-Expendable ICT Equipment
Inverter Aircon (2 HP)	2	35,000	70,000	Semi-Expendable Office Equipment

iPad	1	28,000	28,000	Semi-Expendable ICT Equipment
ASUS Mini PCs	5	25,000	125,000	Semi-Expendable Office Equipment
Smart TV	1	20,000	20,000	Semi-Expendable Office Equipment
Android Tablet	1	15,000	15,000	Semi-Expendable ICT Equipment
Office Security & CCTV	1	15,000	15,000	Semi-Expendable Security Equipment
Android Smartphones	2	12,000	24,000	Semi-Expendable Communication

				Equipment
Heavy-Duty Printer	1	12,000	12,000	Semi-Expendable Printing Equipment
Enterprise Network Infrastructure	1	12,000	12,000	Semi-Expendable ICT Equipment
Flat Monitors	5	6,000	30,000	Semi-Expendable ICT Equipment
Office Tables	15	4,000	60,000	Semi-Expendable Furniture & Fixtures
Ergonomic Chairs	15	3,500	52,500	Semi-Expendable Furniture & Fixtures
Industrial Stand Fans	4	2,500	10,000	Semi-Expendable Office

				Equipment
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This correction transfers PHP 933,500.00 out of the depreciable CapEx, cleaning up the balance sheet for audit compliance.

B. Custom Linux AI Workstation Infrastructure (Ubuntu OS)

To run local AI customer support agents, document verification VLMs, and offline developer copilots, consumer-grade Apple iMacs are both financially inefficient and technically weak.

The founder has completely replaced the original **PHP 390,000.00 Apple iMac budget** with a custom, high-performance Linux-based hardware suite:

Equipment Category	Optimized Linux/Windows Setup Specs (Ubuntu OS)	Unit Cost (PHP)	Quantity	Total Cost (PHP)	Net Savings (PHP)
Central AI Server Workstation	AMD Ryzen 9 7900X (12 Cores), Dual-NVIDIA RTX 3090 (48GB Total	160,000	1	160,000	PHP 35,000

	VRAM), 128GB DDR5 ECC RAM, 2TB NVMe SSD, 1200W Gold PSU				
Developer & Intern Desktops	AMD Ryzen 5 5600 (6 Cores), NVIDIA RTX 3060 12GB VRAM, 32GB DDR4 RAM, 1TB NVMe SSD, 650W Bronze PSU	40,000	3	120,000	PHP 75,000
Total Tech CapEx	—	—	4	PHP 280,000	PHP 110,000

Purpose & Strategic Reallocation of the PHP 110,000.00 Tech Savings:

- The **Central AI Server Workstation** runs **vLLM with continuous batching** and

prefix caching natively on CUDA. This allows it to process up to 50 concurrent customer support chats entirely offline, keeping data completely private and eliminating expensive recurring cloud API costs.

- The **Developer & Intern Desktops** use budget-friendly RTX 3060 12GB cards, allowing junior developers and students to run localized, open-source code autocomplete models (like Qwen2.5-Coder 7B) right on their desks, completely eliminating the need to pay for Copilot subscriptions.
- The **PHP 110,000.00 savings** are reallocated directly to **TODA Partnership Orientations** (PHP 50,000.00) and **Weatherproof Tricycle QR Kits** (PHP 60,000.00), allowing the company to print heavy-duty, laminated QR codes and fund local driver-cooperative onboarding.

C. Recalculating NPV and IRR for Venture Standards

The initial Excel model used a "Firm's Cost of Capital" (discount rate) of **10%**, which resulted in a massive Net Present Value (NPV) of **PHP 25,185,203.71**. For an early-stage, pre-launch software startup, a 10% discount rate is completely unrealistic; it is suited for stable utility bonds, not high-risk tech platforms. Venture capitalists and OrCham investors apply a standard risk-adjusted discount rate of **30% to 40%** to account for the risk of failure.

Applying a realistic venture-grade **30% discount rate** to the Year 1–20 cash flows (comprising a Year 1 Cash Flow of PHP 1,530,000.00, growing at the modeled rate), the recalculation yields:

- **Adjusted Net Present Value (NPV): PHP 1,829,760.00**
- **Internal Rate of Return (IRR): 50.92%** (derived from Year 0 of -PHP 4,019,500.00 and subsequent cash flows).

9. 100-Driver Cohort Launch Simulation (Sustain & Scale Analysis)

To prove operational sustainability from day one, the founder simulated the performance of the **first batch of 100 onboarded drivers** plying the street. Each driver is projected to complete an average of 10 bookings per day (modeled after Tagbilaran's real-world baseline metrics).

A. Monthly Rides-Alone Platform Revenue

- **Total Monthly Bookings:** 100 drivers $\times$ 10 rides/day $\times$ 30 days = **30,000 bookings.**
- **Core Regular Bookings (80%):** 24,000 bookings. At a PHP 15.00 Technology Convenience Fee (F_{tech}), using a conservative **40% Platform Share (PHP 6.00 per ride):**

$$\text{Regular Revenue} = 24,000 \times 6.00 = \text{PHP } 144,000.00$$

- * **Pakyaw Exclusive Bookings (20%):** 6,000 bookings. At a PHP 15.00 Tech Fee, using a **40% Platform Share (PHP 6.00 per ride):**

$$\text{Pakyaw Revenue} = 6,000 \times 6.00 = \text{PHP } 36,000.00$$

- **Total Monthly Rides-Alone Revenue (Model 1 - 40% Platform Share):** PHP 180,000.00.
- **Total Monthly Rides-Alone Revenue (Model 2 - 60% Platform Share):** 30,000 bookings $\times$ PHP 9.00 = **PHP 270,000.00.**

B. Consolidated Monthly Fixed Operational & Manpower Expenses

These figures are taken directly from the initial COSTS.csv spreadsheet and updated with a DOLE-compliant 8-person launch team salary structure:

- **Fixed Operational Expenses (OpEx):**
 - Firebase hosting, optimized mapping APIs, and database syncing: PHP 25,000.00
 - Physical HQ Rental (Prime commercial space): PHP 18,000.00
 - HQ Utilities (LEYECO commercial power for office servers and air cooling): PHP 12,000.00
 - High-speed enterprise fiber business line: PHP 5,000.00
 - Field data allocation and office admin supplies: PHP 3,000.00

- Growth vouchers, local marketing, and driver peak-hour incentives: PHP 108,333.34
- *Total Fixed OpEx: **PHP 171,333.34***
- **Launch Team Salaries (8-Person Payroll):**
 - Project Lead (CEO): PHP 45,000.00
 - Lead Developer (CTO): PHP 45,000.00
 - Junior Developer/Tester: PHP 25,000.00
 - Operations & Finance Officer: PHP 30,000.00
 - Marketing & Support Lead: PHP 25,000.00
 - In-Office Support Agents (2): PHP 30,000.00 (PHP 15,000.00 each)
 - DOLE-SPES Student Interns (2): PHP 20,000.00 (PHP 10,000.00 each - private employer's 60% counterpart wage under RA 10917)
 - *Total Manpower Salaries: **PHP 220,000.00***
- **Total Monthly Operational Burn (Consolidated): **PHP 391,333.34****

C. The Deficit Gap & B2B SaaS Solution (The Investor Hole)

- **The Problem:** Under a "Rides-Alone" model, the first batch of 100 drivers generates only PHP 180,000.00 (40% split) or PHP 270,000.00 (60% split) in monthly revenue. This leaves a monthly deficit of **PHP 121,333.34 to PHP 211,333.34** against the heavy PHP 391K operational burn. If pitched to OrCham, they will reject it as unsustainable.
- **The Solution:** The founder must activate the high-margin B2B SaaS channels immediately upon launching the first batch:
 1. **"Pakyaw Perks" B2B Merchant Referrals:** Sourcing PHP 30.00 lead-generation fees from local commercial partners (SM, Robinsons, Sabin) for delivering commuters. Delivering just 150 customers per day to these establishments generates:

$$150 \text{ referrals/day} \times 30.00 \times 30 \text{ days} = \text{PHP } 135,000.00 \text{ monthly revenue}$$

2. **B2B Hyperlocal Logistics ("Pasabuy" / "Pasugo"):** Repurposing idle driver off-peak hours (10:00 AM to 3:00 PM) to complete 800 monthly deliveries for local merchants, pharmacies, and laundry shops at a PHP 40.00 platform cut:

$$800 \text{ deliveries/month} \times 40.00 = \text{PHP } 32,000.00 \text{ monthly revenue}$$

3. **SaaS Fleet Owner Subscriptions:** Onboarding 10 local fleet operators managing blocks of tricycles, charging a PHP 499.00 monthly subscription for tracking and driver shifts:

$$10 \text{ fleets} \times 499.00 = \text{PHP } 4,990.00 \text{ monthly revenue}$$

D. Consolidated Month 1 Sustainability Balance Sheet

By combining the rides revenue with the high-margin B2B SaaS channels, the cash flow balance is evaluated:

Financial Metric	Model 1 (40% Platform Share)	Model 2 (60% Platform Share)
Rides Platform Revenue	PHP 180,000.00	PHP 270,000.00
B2B Merchant Referrals	PHP 135,000.00	PHP 135,000.00
B2B Hyperlocal Logistics	PHP 32,000.00	PHP 32,000.00
SaaS Fleet Subscriptions	PHP 4,990.00	PHP 4,990.00
Total Monthly Revenue	PHP 351,990.00	PHP 441,990.00
<i>Less: Operating Expenses (OpEx)</i>	(PHP 171,333.34)	(PHP 171,333.34)

<i>Less: Manpower Salaries (8-Pax)</i>	(PHP 220,000.00)	(PHP 220,000.00)
Net Cash Flow (Month 1)	-PHP 39,343.34 (Sustained by Runway)	+PHP 50,656.66 (Net Profitable)

In Model 2 (60% platform share), the first batch of 100 drivers successfully achieves **net profitability (+PHP 50,656.66) from Month 1**, proving the platform is a robust, self-sustaining financial powerhouse.

10. Cancellation Governance Framework: Eliminating Ghost Bookings, Driver Abandonment, and Marketplace Manipulation

A transportation marketplace cannot rely solely on successful bookings. Real-world operations inevitably involve booking cancellations arising from passenger emergencies, accidental requests, driver vehicle breakdowns, weather disturbances, and intentional bad-faith actors.

Without structured cancellation governance, the platform becomes vulnerable to:

- Ghost bookings that waste driver fuel and time.
- Drivers accepting rides only to abandon passengers after receiving higher-paying opportunities.
- Artificial supply shortages caused by repeated cancellations.
- Reduced customer confidence and lower platform retention.

To preserve marketplace integrity while maintaining fairness to both parties, Pakyaw implements a dual-sided Cancellation Governance Framework.

A. Passenger-Initiated Cancellation Heuristic

Passenger cancellations are evaluated according to the driver's operational commitment level.

Scenario 1: Immediate Passenger Cancellation

A passenger accidentally books the wrong destination and cancels within sixty (60) seconds of submitting the booking request.

Operational Conditions:

- Driver has not yet traveled toward the pickup location.
- No significant fuel expenditure has occurred.

Platform Resolution:

- Ride automatically cancelled.
- Passenger charged PHP 0.00.
- Driver receives no compensation.

This creates a correction window for genuine user mistakes while preventing unnecessary support requests.

Scenario 2: Driver Already En Route

A passenger books a Solo Pakyaw ride.

The driver accepts and travels approximately 1.5 kilometers toward the pickup location.

The passenger suddenly cancels before driver arrival.

Operational Impact:

- Fuel already consumed.
- Driver loses alternative booking opportunities.
- Driver occupancy time wasted.

Platform Resolution:

Passenger Cancellation Fee:

PHP 10.00

Revenue Allocation:

- Driver Compensation: PHP 8.00
- Platform Administrative Fee: PHP 2.00

This transforms fuel loss into a recoverable operational expense rather than a driver subsidy.

Scenario 3: Passenger No-Show

The driver successfully arrives at the designated pickup point.

The passenger fails to board.

The application automatically activates a five-minute countdown timer.

After timer expiration:

Ride Status:

"No Passenger Present"

Platform Resolution:

Ride Type	No-Show Fee
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Share Ride	PHP 10.00
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Solo Pakyaw	PHP 20.00
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Revenue Distribution:

- 80% Driver
- 20% Platform

This discourages deliberate no-shows while compensating drivers for lost productivity.

B. Driver-Initiated Cancellation Governance

Unlike passengers, drivers directly influence marketplace reliability after accepting a booking. Therefore, stricter accountability standards are applied.

Scenario 4: Accidental Driver Acceptance

A driver accidentally taps the Accept button.

Cancellation occurs within thirty (30) seconds.

Platform Resolution:

- No strike issued.
- No financial penalty.
- Booking returned to dispatch queue.

This accommodates normal human error.

Scenario 5: Driver Cancels While En Route

A driver accepts a booking.

The driver begins traveling toward the passenger.

The driver subsequently cancels after receiving a higher-paying street passenger.

Operational Impact:

- Passenger waiting time increases.
- Marketplace trust decreases.
- Artificial supply distortion occurs.

Platform Resolution:

- Automatic Driver Strike recorded.
- **Passenger receives priority rematching.**
- Driver reliability score reduced.

Repeated occurrences affect future dispatch priority.

Scenario 6: Chronic Driver Cancellation Behavior

To identify unreliable operators, the platform continuously monitors cancellation percentages.

Monthly Driver Cancellation Thresholds:

Cancellation Rate Enforcement Action

Below 5%	No Action
5% - 10%	Warning Notice
10% - 15%	Incentive Reduction
Above 15%	Temporary Suspension

This creates a measurable accountability structure without unfairly penalizing legitimate emergencies.

C. Shared Ride Cancellation Safeguards

Share mode introduces additional complexity because multiple passengers depend on a common vehicle dispatch.

Scenario 7: Passenger Cancels Before Occupancy Lock

The platform requires three matched passengers before dispatching a vehicle.

Passenger A, Passenger B, and Passenger C are currently queued.

Passenger B cancels.

Platform Resolution:

- Driver not dispatched.
- Remaining passengers remain in batching queue.
- No cancellation fee assessed.

The algorithm simply searches for a replacement passenger.

Scenario 8: Occupancy Threshold Collapse

Three passengers successfully match.

One passenger cancels immediately before dispatch.

Occupancy falls below the minimum threshold.

Platform Resolution:

Remaining passengers receive:

"Your Share route no longer satisfies dispatch requirements."

Passenger Options:

1. Continue waiting for additional riders.
2. Upgrade to Solo Pakyaw.
3. Cancel booking without penalty.

This preserves user flexibility while preventing platform cash bleed.

D. Dynamic Cancellation Risk Pricing

Periods of abnormal demand create significantly higher driver opportunity costs.

Examples:

- Heavy Rainfall
- Typhoon Conditions
- Fiesta Events
- Rush-Hour Congestion
- Holiday Traffic Surges

During these periods, cancellation penalties may increase by PHP 5.00 to PHP 10.00.

Maximum cancellation fee:

PHP 30.00

This discourages abuse during periods where driver availability is critically constrained.

11. Trust, Safety, and Dispute Resolution Architecture

For a transport marketplace to scale beyond the founder's direct supervision, operational trust must be embedded into the platform architecture itself.

Pakyaw therefore establishes a multi-layer Trust, Safety, and Dispute Resolution System protecting both passengers and drivers.

A. Passenger-to-Driver Incident Reporting

Passengers may submit reports for up to seventy-two (72) hours after ride completion.

Each report automatically generates a digital case file containing:

- GPS route history
 - Pickup coordinates
 - Drop-off coordinates
 - Ride timestamps
 - Driver identification
 - Passenger statement
-

Scenario 1: Fare Manipulation Attempt

Official platform fare:

PHP 16.00

Driver demands:

PHP 30.00 cash payment.

Passenger submits report.

Platform Investigation:

Evidence Sources:

- Fare receipt
- GPS logs
- Booking records

Enforcement Ladder:

First Offense:

- Formal Warning

Second Offense:

- Seven-Day Suspension

Third Offense:

- Permanent Platform Removal
-

Scenario 2: Reckless Driving Complaint

A passenger reports:

- Excessive speeding
- Dangerous overtaking
- Traffic law violations

Evidence Submitted:

- Dashcam footage
- Mobile phone recording
- Passenger testimony

Platform Resolution:

- Safety review initiated.
 - Driver temporarily restricted if risk level is elevated.
 - Mandatory safety orientation may be required.
-

Scenario 3: Harassment or Passenger Intimidation

Examples:

- Verbal abuse

- Threatening language
- Discriminatory remarks
- Physical intimidation

Platform Resolution:

Account immediately enters:

"Safety Review Status"

Possible Outcomes:

- Warning
 - Temporary Suspension
 - Permanent Ban
 - Referral to local authorities when legally necessary
-

B. Driver-to-Passenger Incident Reporting

Trust and safety protections apply equally to drivers.

Scenario 4: Passenger Refuses Payment

Passenger completes journey.

Passenger exits vehicle without paying.

Driver submits incident report.

Platform Investigation:

Evidence Sources:

- Ride completion logs
- GPS verification
- Driver statement

Enforcement:

- Passenger account flagged.
- Outstanding balance attached to account.
- Repeat offenders permanently removed.

Scenario 5: Passenger Harasses Driver

Examples:

- Threats
- Verbal abuse
- Disorderly conduct

Platform Resolution:

Trust & Safety Review initiated.

Possible Outcomes:

- Warning
 - Temporary Suspension
 - Permanent Account Ban
-

Scenario 6: Passenger Damages Vehicle Property

A passenger intentionally damages:

- Vehicle seat
- Sidecar equipment
- Laminated QR Code System
- Safety signage

Driver uploads photographic evidence.

Platform Resolution:

- Investigation initiated.
- Passenger may be required to reimburse verified damages.

- Repeat offenders permanently removed.
-

C. Emergency Response and Active Safety Monitoring

To support passenger and driver welfare, Pakyaw integrates an Emergency Safety Layer.

One-Tap SOS Function

Available to:

- Drivers
- Passengers

Upon activation:

- Live GPS coordinates transmitted.
- Emergency incident ticket generated.
- Operations dashboard alerted.
- Emergency contact notified.

This provides immediate escalation capability during safety-critical situations.

D. Reputation-Based Marketplace Governance

Every completed ride contributes to a continuously updated trust score.

Passenger Trust Score

Evaluation Factors:

- No-show frequency
- Payment disputes
- Cancellation behavior
- Verified complaints

Range:

0–100

Driver Trust Score

Evaluation Factors:

- Acceptance rate
- Cancellation rate
- Safety incidents
- Passenger ratings

Range:

0–100

Marketplace Reputation Tiers

Score Range Classification

90-100	Gold
75-89	Silver
60-74	Standard
Below 60	At-Risk

Higher trust scores receive:

- Priority matching.
- Promotional incentives.
- Faster customer support resolution.

E. Fraud Detection and Abuse Prevention Layer

To protect platform integrity, the central AI server continuously monitors abnormal behavior patterns.

The system automatically flags:

- Repeated false reporting.
- Excessive cancellations.

- GPS inconsistencies.
- Multi-account creation attempts.
- Coordinated fare manipulation.
- Suspicious ride completion anomalies.

Flagged accounts are routed to manual review before enforcement actions are applied.

Real-World Operational Example

A passenger books a Solo Pakyaw ride during evening peak hours.

The driver accepts and begins traveling toward the pickup point.

The passenger cancels after four minutes.

The system automatically charges the applicable cancellation fee and compensates the driver for fuel expenditure.

Conversely, if the driver cancels after accepting the booking in order to pursue an off-app cash-paying passenger, the driver receives a reliability strike, reduced dispatch priority, and possible suspension upon repeated violations.

Through this framework, Pakyaw transforms cancellations, disputes, and misconduct from unmanaged operational risks into measurable, enforceable, and scalable governance processes.